

Two independent standards validate evidence recovery and penetrance calibration

Manual adjudication tests extraction; held-out biobank carriers test generalization

Gold standard 1 – Frozen manual literature adjudication
Was the literature recovered correctly?

Calibration standard 2 – Independent biobank carrier outcomes
Does the penetrance estimate generalize?

A Evidence retained



Discovery miss
Parsing failure

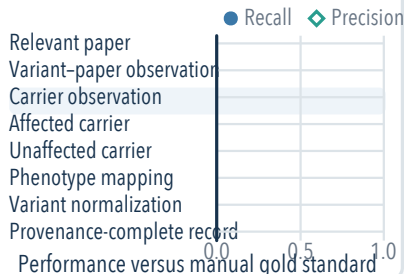
Triage false negative
Normalization failure

Access failure
Incomplete provenance

Discovery and access failures are scored separately from extraction and provenance failures.

B Extraction performance

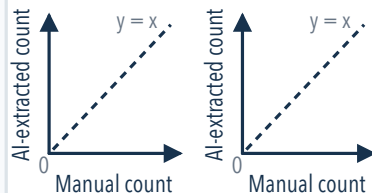
Primary endpoint: carrier-level performance



C Carrier-count agreement

Scored at variant-cohort-phenotype level.

Affected carriers Unaffected carriers



MAE · Median absolute error
Exact match · Within ± 1 carrier

D Failure modes

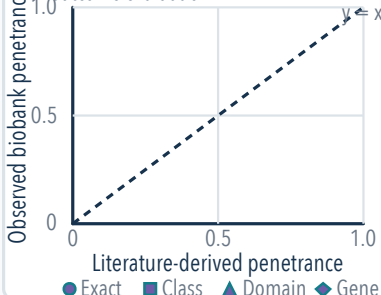
- Discovery miss
- Triage false negative
- Inaccessible full text
- Supplement/table parsing
- Phenotype mapping
- Variant normalization
- Cohort-total leakage
- Duplicate cohort
- Incomplete provenance
- Other adjudicated error

Adjudicated errors

High-impact errors trigger mandatory human review.

E External calibration

🔒 Literature model frozen before outcome evaluation



F Supported biological resolution

Resolution	Cal. error ← Better	Interval coverage 50 · 80 · 95%	Estimate coverage
R0 Gene / mechanism			
R1 Consequence class			
R2 Domain / region			
R3 Structure / dynamics			
R4 Leakage-safe neighborhood			

Select resolution only after held-out calibration.

Accurate evidence recovery is necessary; independent calibration determines whether the estimate generalizes.